E X E R C I S E 0 2 · S Q L FOUNDATIONS SQL Aggregate Functions & Operators

QUESTION 01: List all distinct departments in the students table. EXPECTED COLUMNS: department

```
22, 'HR');

SELECT * FROM BUCCLEUCH.PRIMARY.LEARNERS;
--select distinct department from BUCCLEUCH.PRIMARY.LEARNERS
SELECT DISTINCT department FROM BUCCLEUCH.PRIMARY.LEARNERS;
```

Add parameter

Table

	department
1	IT
2	HR
3	Finance

Question 02: Get the average age of students per department. EXPECTED COLUMNS: department avg_age

```
--Get the average age of students per department
SELECT department, AVG (age) AS average_age FROM BUCCLEUCH.PRIMARY.LEARNERS GROUP BY department;
```

Add parameter

Table

	department	1.2 average_age
1	IT	20.5
2	HR	22
3	Finance	23

Question 03: Show departments with more than 1 student. EXPECTED COLUMNS: department student_count

```
13  --Show departments with more than 1 student.dep and student
14  SELECT department, COUNT(student_id) AS student_count FROM BUCCLEUCH.PRIMARY.LEARNERS GROUP BY department HAVING student_count > 1;
```

Add parameter

Table

	department	student_count
	IT	2
	HR	2

Question 04:Get all students whose age is between 21 and 23. EXPECTED COLUMNS: student_id name age department

```
16  --Get all students whose age is between 21 and 23.
17  SELECT student_id, name, age, department FROM BUCCLEUCH.PRIMARY.LEARNERS WHERE age BETWEEN 21 AND 23;
```

Add parameter

Table

	student_id	name	age	department
1	2	Bob	22	HR
2	3	Charlie	21	IT
3	4	Diana	23	Finance
4	5	Eve	22	HR

Question 05:List all students in the IT or HR department who are older than 21. EXPECTED COLUMNS: student_id name age department

```
19  --List all students in the IT or HR department who are older than 21.
20  SELECT student_id, name, age,department FROM BUCCLEUCH.PRIMARY.LEARNERS WHERE department IN ( 'IT', 'HR') AND age > 21;
```

Add parameter

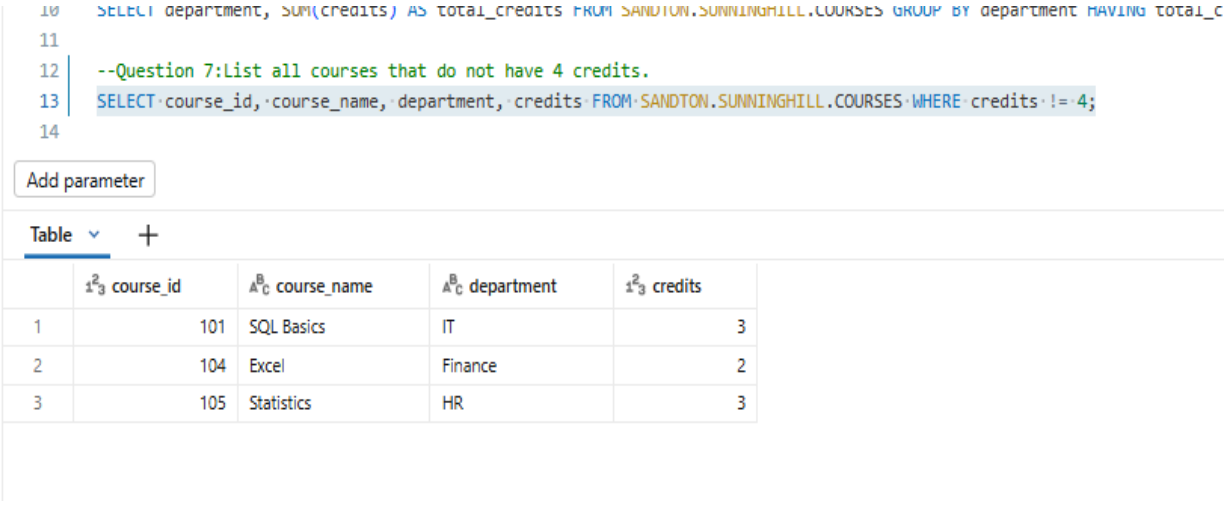
Table

	student_id	name	age	department
1	2	Bob	22	HR
2	5	Eve	22	HR

Question 06:Show total credits per department, only for departments with more than 5 total credits. EXPECTED COLUMNS: department total_credits



Question 07: List all courses that do not have 4 credits. EXPECTED COLUMNS: course_id course_name department credits



Question 08: Show the top 3 courses by credits in descending order. EXPECTED COLUMNS: course_id course_name credits

```
--Question 8: Show the top 3 courses by credits in descending order.
SELECT course_id, course_name, credits FROM SANDTON.SUNNINGHILL.COURSES ORDER BY credits DESC LIMIT 3;
```

Add parameter

Table

	course_id	course_name	credits
1	102	Python	4
2	103	Data Science	4
3	101	SQL Basics	3

Question 9:Get the maximum, minimum, and average grade across all enrollments. EXPECTED COLUMNS: max_grade min_grade avg_grade

```
Create CATALOG if not exists mohodi;
Create SCHEMA if not exists mohodi.fatima;
Create TABLE if not exists mohodi.fatima.enrollments (enrollment_id int,student_id int,course_id int,grade int);
INSERT INTO mohodi.fatima.enrollments VALUES (1,1,101,85),(2,2,102,78),(3,3,103,90),(4,4,104,88),(5,5,105,82);

SELECT * FROM mohodi.fatima.enrollments;
--Question 9:Get the maximum, minimum, and average grade across all enrollments
SELECT MAX(grade) AS max_grade,MIN(grade) AS min_grade, AVG(grade) AS avg_grade FROM mohodi.fatima.enrollments;
```

Add parameter

Table

	max_grade	min_grade	avg_grade
1	90	78	84.6

Question 10: Count how many enrollments exist per course. EXPECTED COLUMNS: course_id enrollment_count

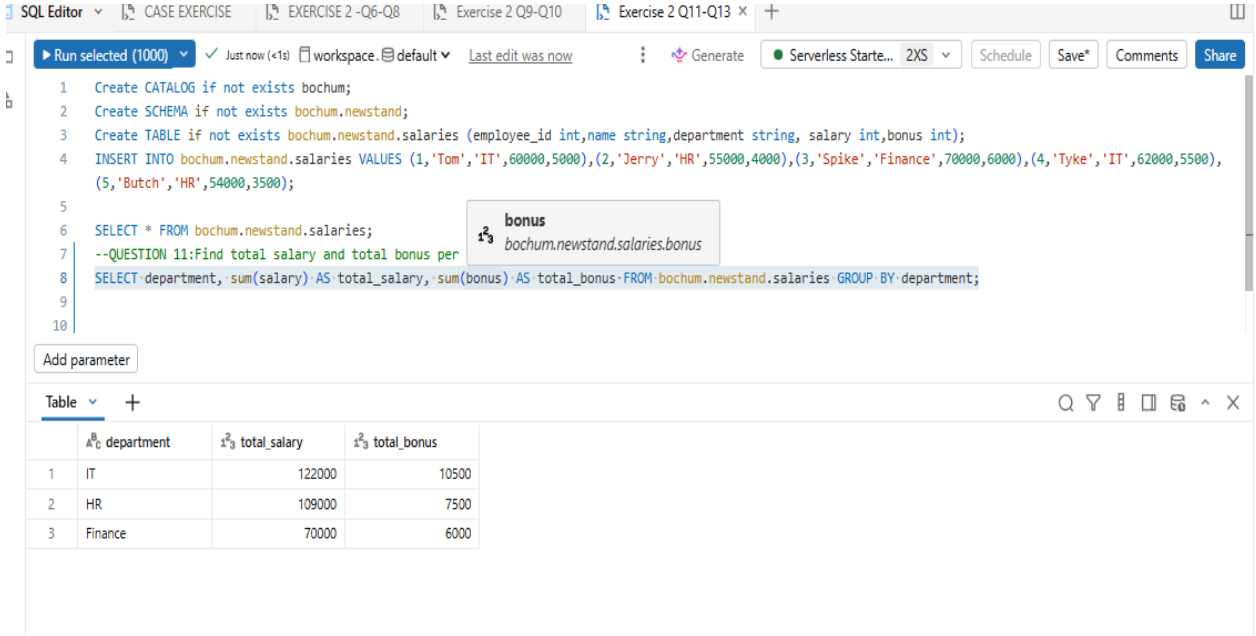
```
--Question 10: Count how many enrollments exist per course.
SELECT course_id,COUNT(enrollment_id) AS enrollment_count FROM mohodi.fatima.enrollments GROUP BY course_id;
```

Add parameter

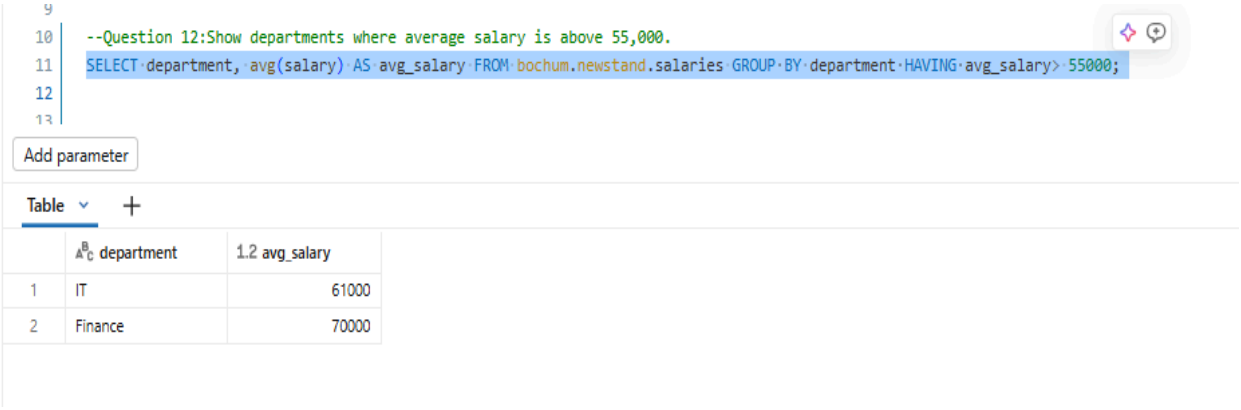
Table

	course_id	enrollment_count
1	101	1
2	102	1
3	103	1
4	104	1
5	105	1

Question 11: Find total salary and total bonus per department. EXPECTED COLUMNS: department total_salary total_bonus



Question 12: Show departments where average salary is above 55,000. EXPECTED COLUMNS: department avg_salary



Question 13: List employees whose salary plus bonus is greater than 60,000. EXPECTED COLUMNS: employee_id name salary bonus total_compensation

```
12
13  --Question 13:List employees whose salary plus bonus is greater than 60,000.
14  SELECT employee_id,name,salary,bonus, sum(salary+bonus) AS total_compensation FROM bochum.newstand.salaries GROUP BY
    employee_id,name,salary,bonus HAVING total_compensation>60000;
15
16
```

Add parameter

Table

	employee_id	name	salary	bonus	total_compensation
1	1	Tom	60000	5000	65000
2	3	Spike	70000	6000	76000
3	4	Tyke	62000	5500	67500

Question 14: Show total and average budget per department. Only include departments with average budget above 70,000. EXPECTED COLUMNS: department total_budget avg_budget

```
1  Create CATALOG if not exists polokwane;
2  Create SCHEMA if not exists polokwane.moletjie;
3  Create TABLE if not exists polokwane.moletjie.projects (project_id INT, project_name STRING, department STRING, budget int);
4  INSERT INTO polokwane.moletjie.projects  VALUES (1, 'AI App', 'IT', 120000), (2, 'Payroll System', 'Finance', 80000), (3, 'Dashboard',
   'IT', 150000), (4, 'Website', 'Marketing', 60000),(5, 'HR Portal', 'HR', 50000);
5
6  SELECT * FROM polokwane.moletjie.projects;
7  --Question 14: Show total and average budget per department. Only include departments with average budget
8  above 70,000
9  SELECT department, SUM(budget) as total_budget, AVG(budget) as avg_budget FROM polokwane.moletjie.projects GROUP BY department HAVING
   avg_budget > 70000;
```

Add parameter

Table

	department	total_budget	avg_budget
1	IT	270000	135000
2	Finance	80000	80000

Question 15: List all projects with budgets between 50,000 and 120,000, excluding the Marketing department. EXPECTED COLUMNS: project_id project_name department budget

```
10
11  --Question 15: List all projects with budgets between 50,000 and 120,000, excluding the Marketing department
12  SELECT project_id, project_name, department, budget FROM polokwane.moletjie.projects WHERE budget BETWEEN 50000 AND 120000 AND department
    != 'Marketing';
```

Add parameter

Table

	project_id	project_name	department	budget
1	1	AI App	IT	120000
2	2	Payroll System	Finance	80000
3	5	HR Portal	HR	50000